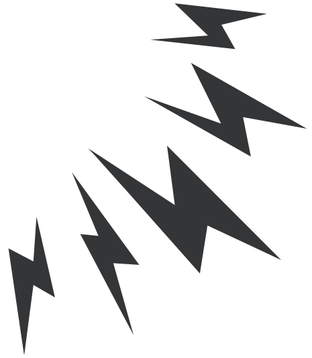


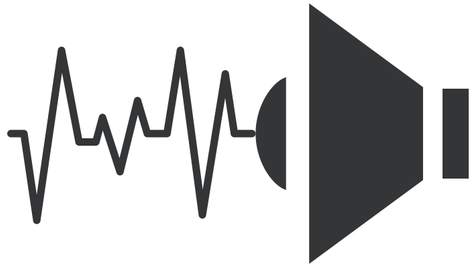


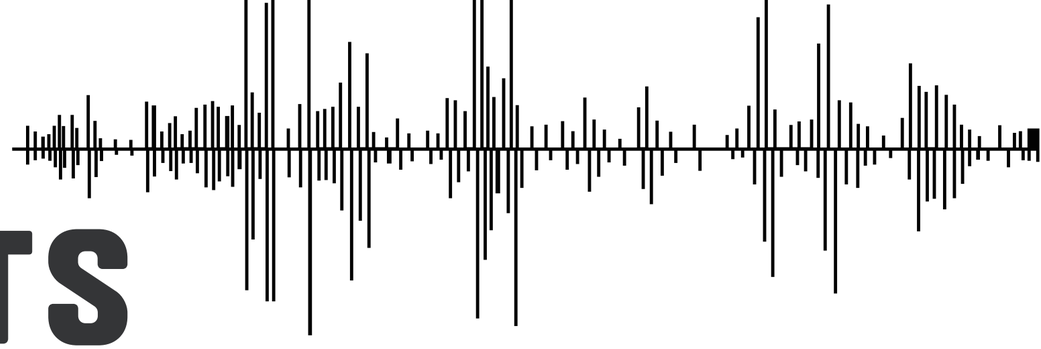
# AUDIO NOISE REDUCTION

Digital Signal Processing - Term Project

Group 02





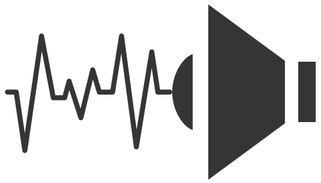


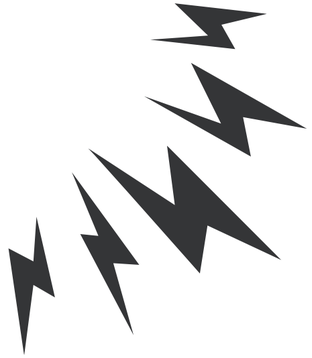
# CONTENTS


- Overview
- Audio Preparation
- Exploratory Data Analysis
- Noise reduction pipeline
- Test and validate algorithm with noisy records
- Failure Cases
- Conclusion





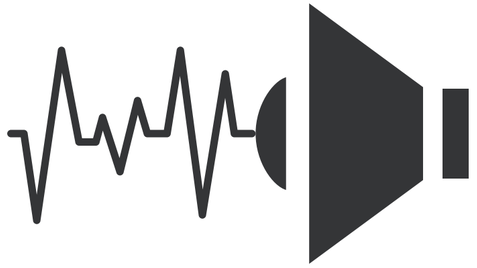







# OVERVIEW



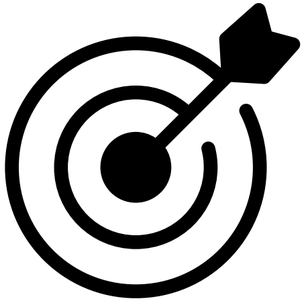


# OVERVIEW



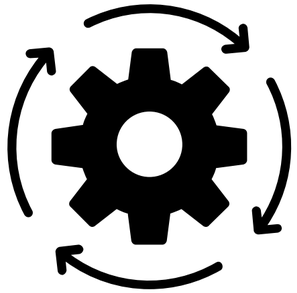


1



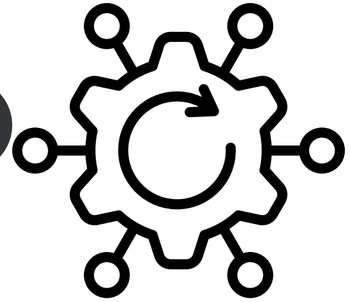
Restore real world noisy audio by DSP algorithm

2



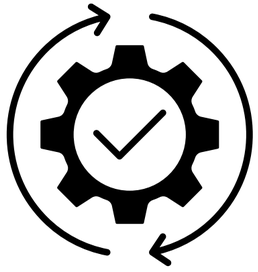
Add custom noise profiles with clean reference audio

3



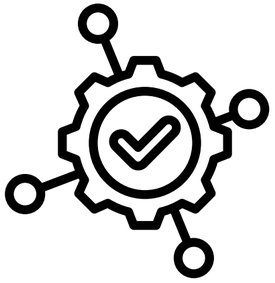
Build DSP pipeline based on the synthetic baseline

4



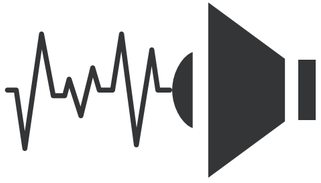
Test the algorithm against unknown noisy records

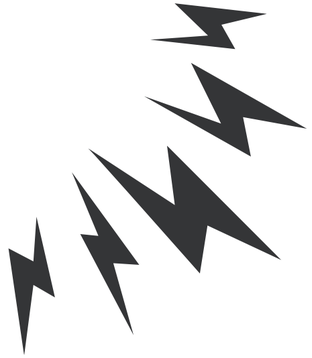
5



Use algorithm with real world noisy audio



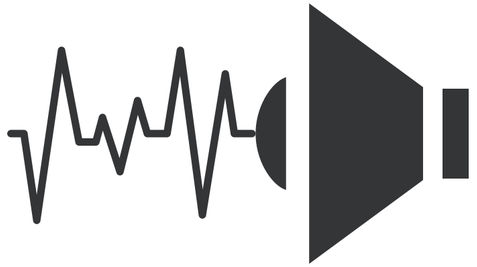




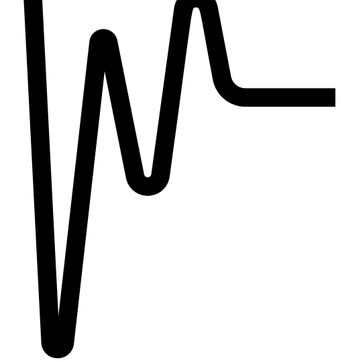


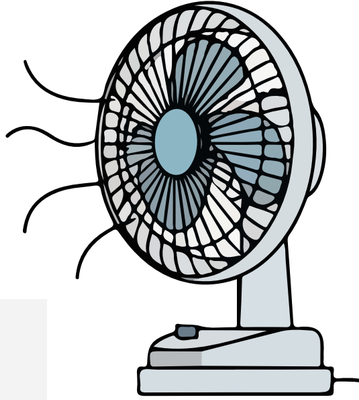
# AUDIO PREPARATION





# AUDIO PREPARATION



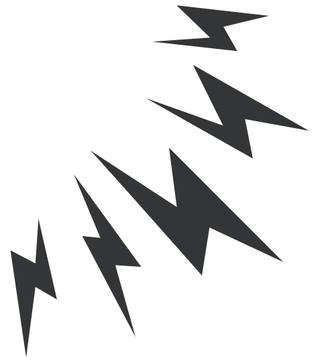


- audio
  - clean_speech.wav
  - hci_audio_record.m4a
  - hci_audio_record.wav
  - noise_recording.m4a
  - noise_recording.wav

- noisy_audio
  - noisy_1.wav
  - noisy_2.wav
  - noisy_3.wav
  - noisy_4.wav
  - noisy_5.wav
  - noisy_6.wav
  - noisy_7.wav
  - noisy_8.wav
  - noisy_9.wav
  - noisy_10.wav



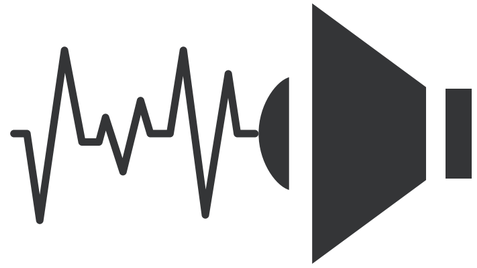






# EXPLORATORY DATA ANALYSIS (EDA)





# EXPLORATORY DATA ANALYSIS



**Convert from .m4a file to .wav file**

!ffmpeg -y -loglevel error -i audio/sound_1.m4a -ac 1 -ar 44100 -c:a pcm_s16le audio/sound_1.wav

### clean_speech.wav

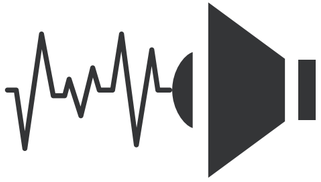
```
fs = 16000
Samples = 172343
Duration = 10.7714375 s
signal shape = (172343,)
signal dtype = int16
```

### noise_recording.wav

```
fs = 44100
Samples = 587001
Duration = 13.310680272108844 s
signal shape = (587001,)
signal dtype = int16
```

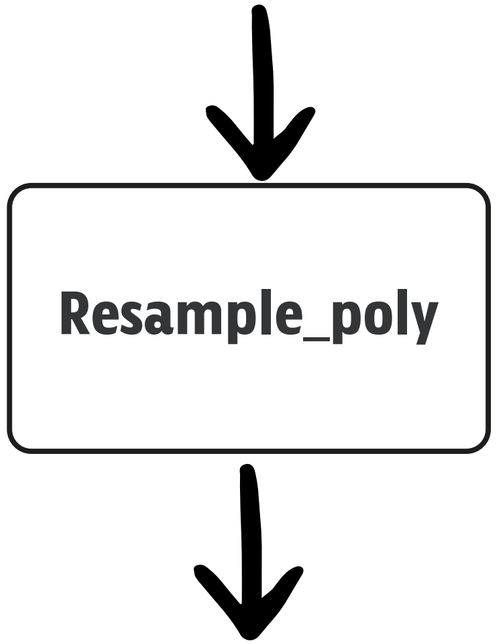


```
1   noise = noise_signal.astype(np.float64)
2   noise = noise / np.max(np.abs(noise))
```



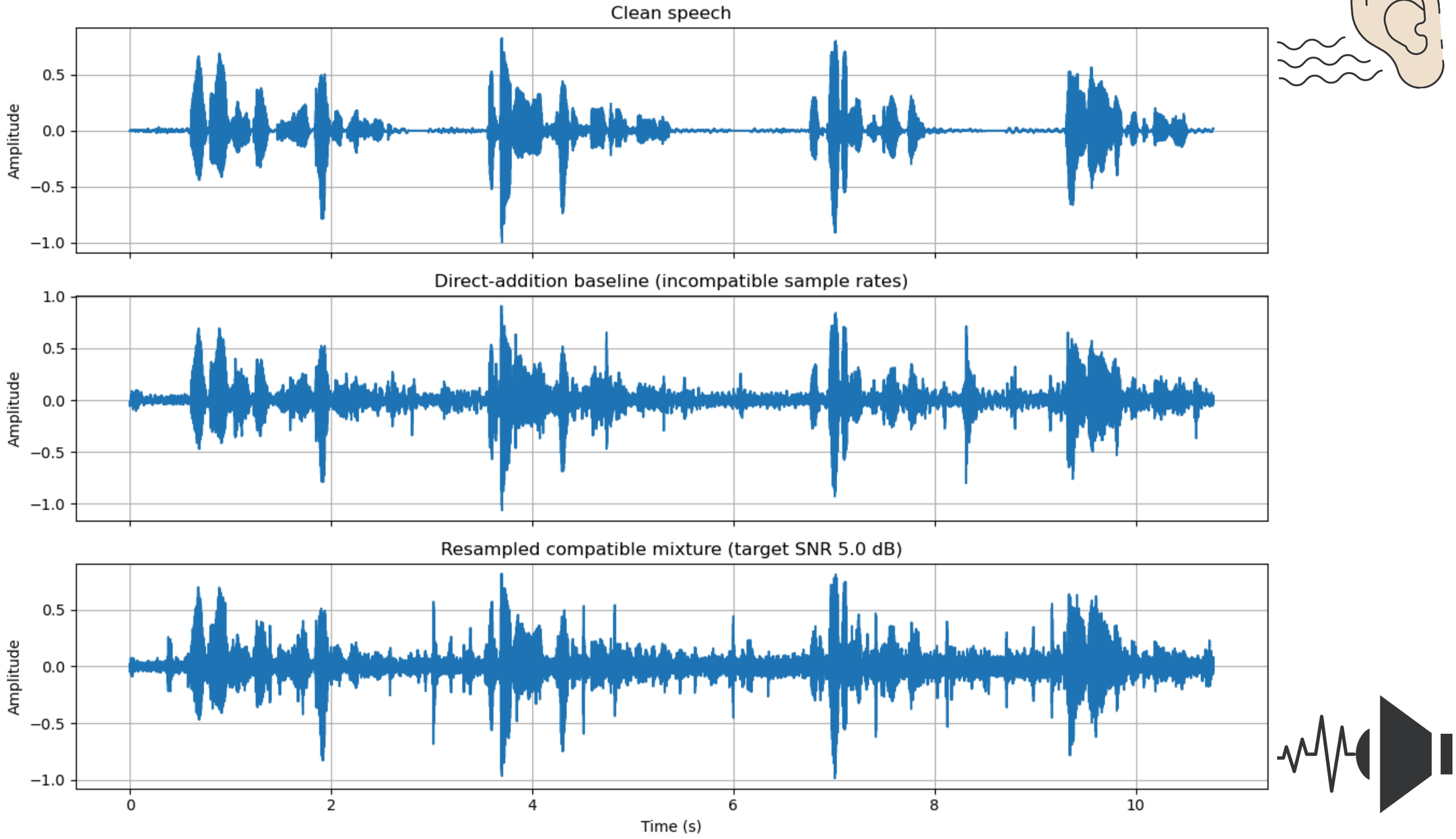
# EXPLORATORY DATA ANALYSIS

```
Clean speech: 16000 Hz, 172343 samples (10.77 s)
Noise:        44100 Hz, 587001 samples (13.31 s)
```



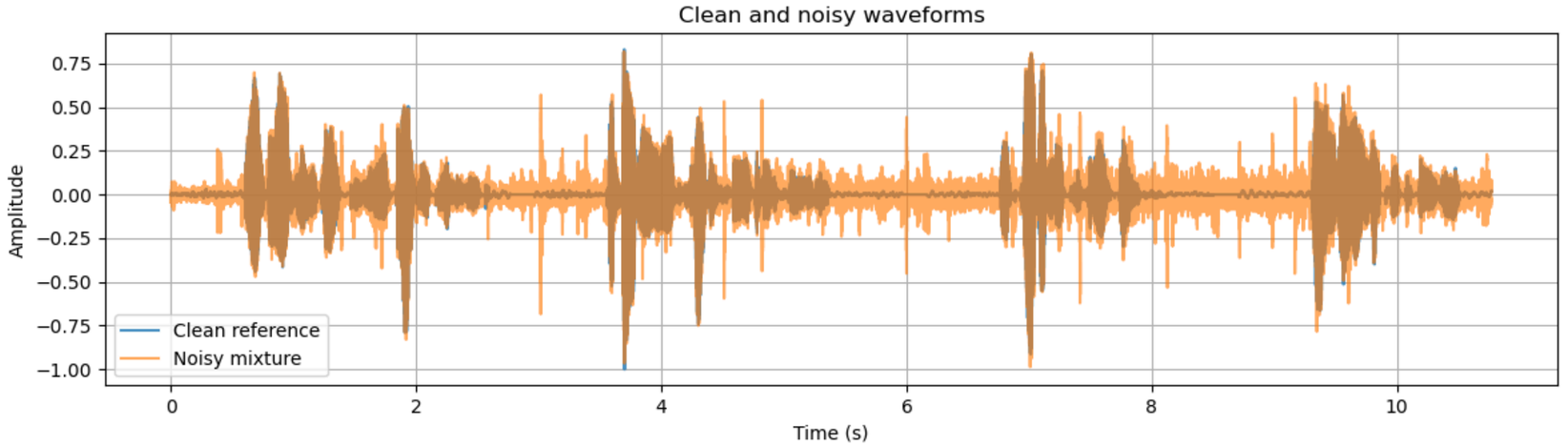

```
Noise sampling rate: 44100 Hz -> 16000 Hz
Baseline samples: 172343
Compatible samples: 172343
Baseline mixture SNR estimate: 5.16 dB
Compatible mixture SNR: 5.00 dB (target 5.00 dB)
```

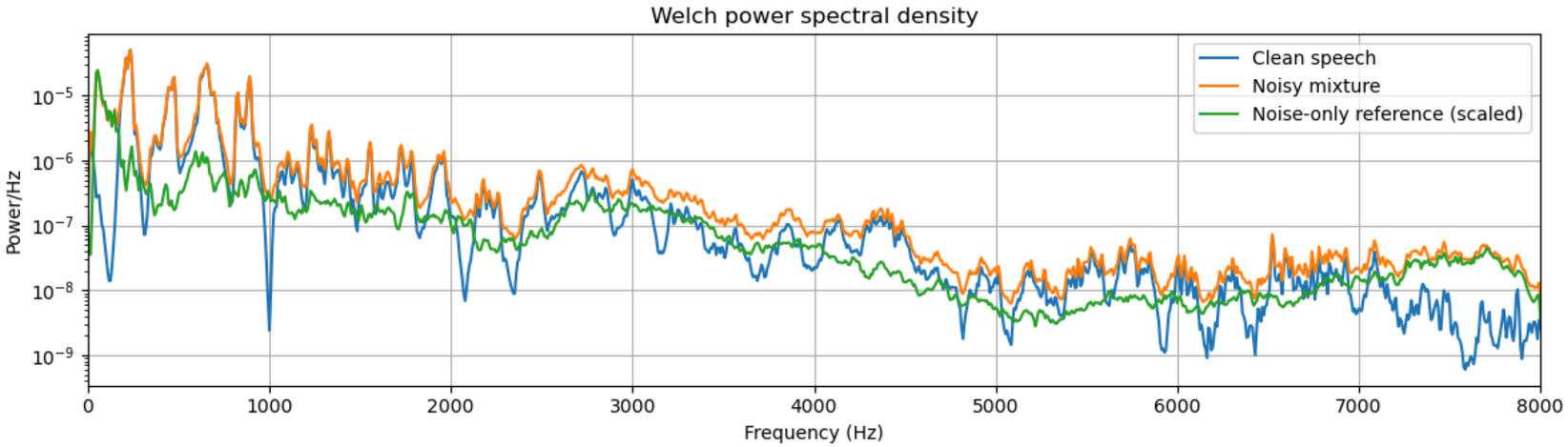
# EXPLORATORY DATA ANALYSIS

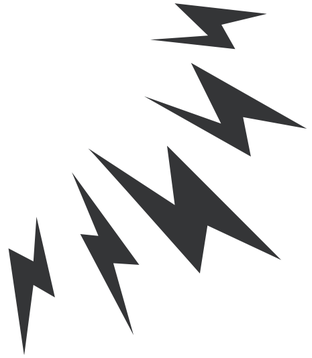


# EXPLORATORY DATA ANALYSIS

```
Noise PSD power by candidate filter band:
  0-70 Hz: 0.000510092 (24.32% of estimated noise power)
  70-7500 Hz: 0.00145155 (69.21% of estimated noise power)
  7500-8000 Hz: 1.21106e-05 (0.58% of estimated noise power)
```



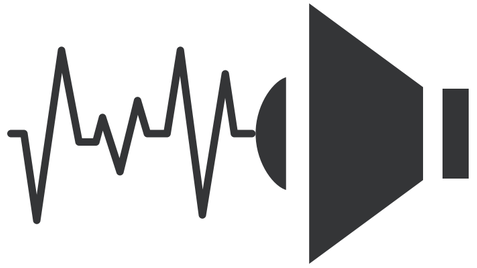







# NOISE REDUCTION PIPELINE



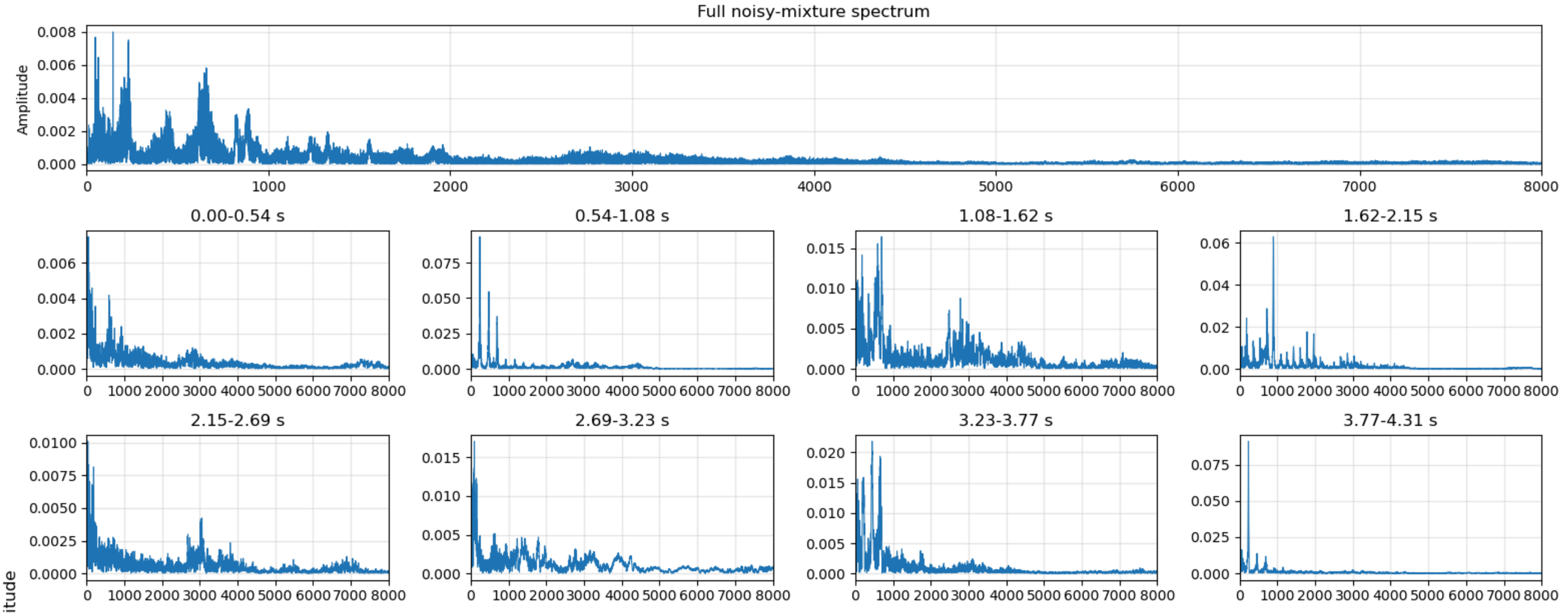


# FFT SPECTRUM





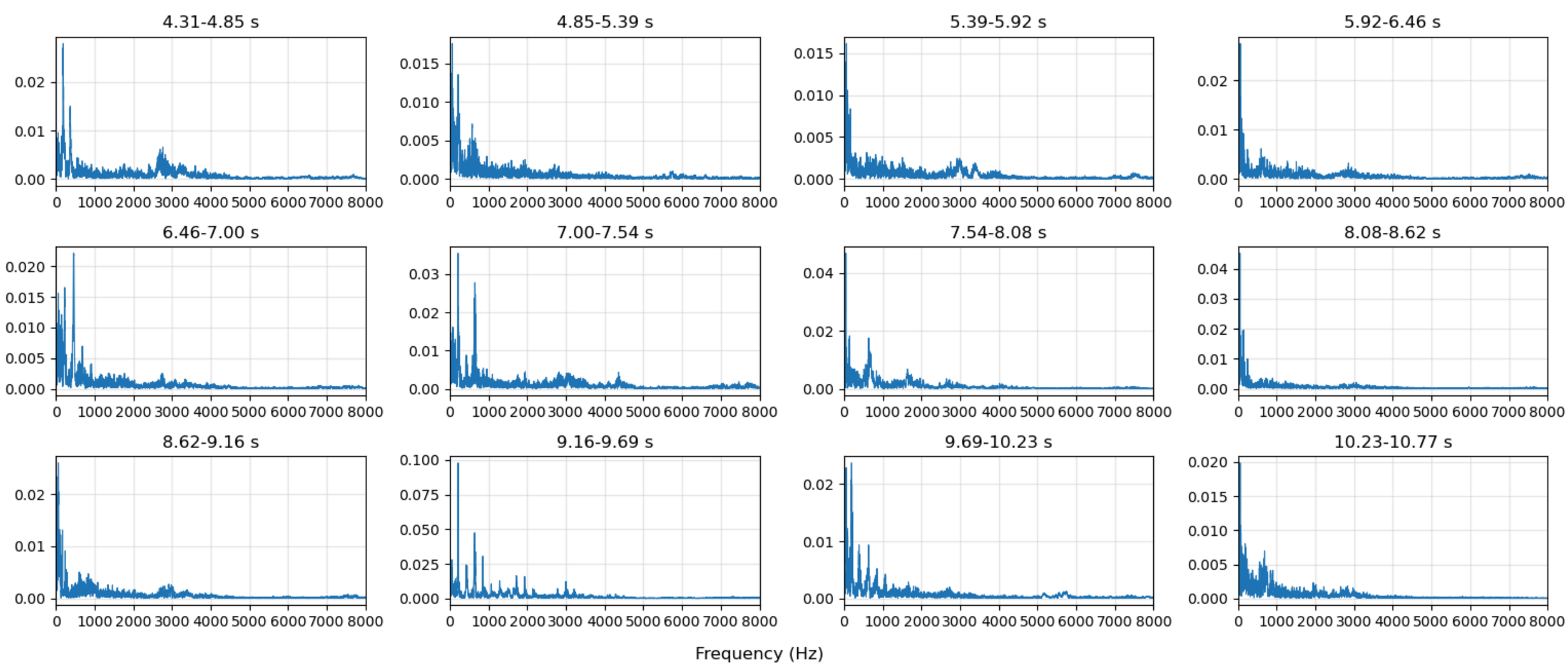
Noisy mixture full spectrum and 20 time frames



# FFT SPECTRUM

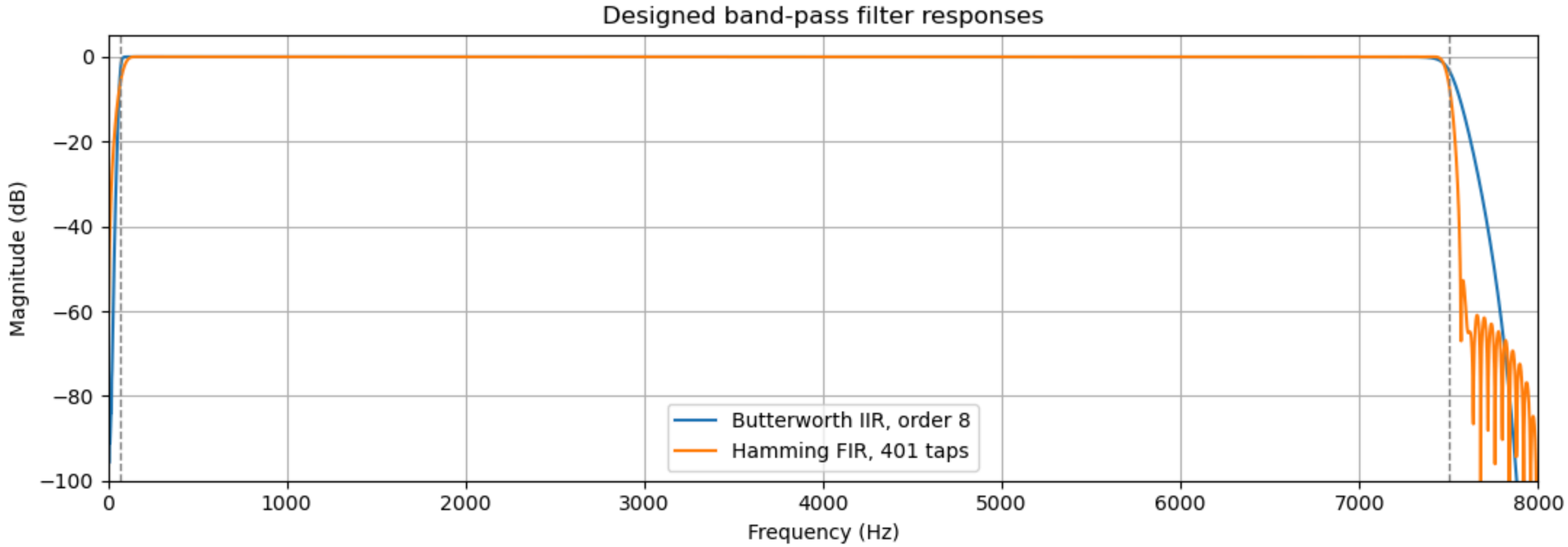




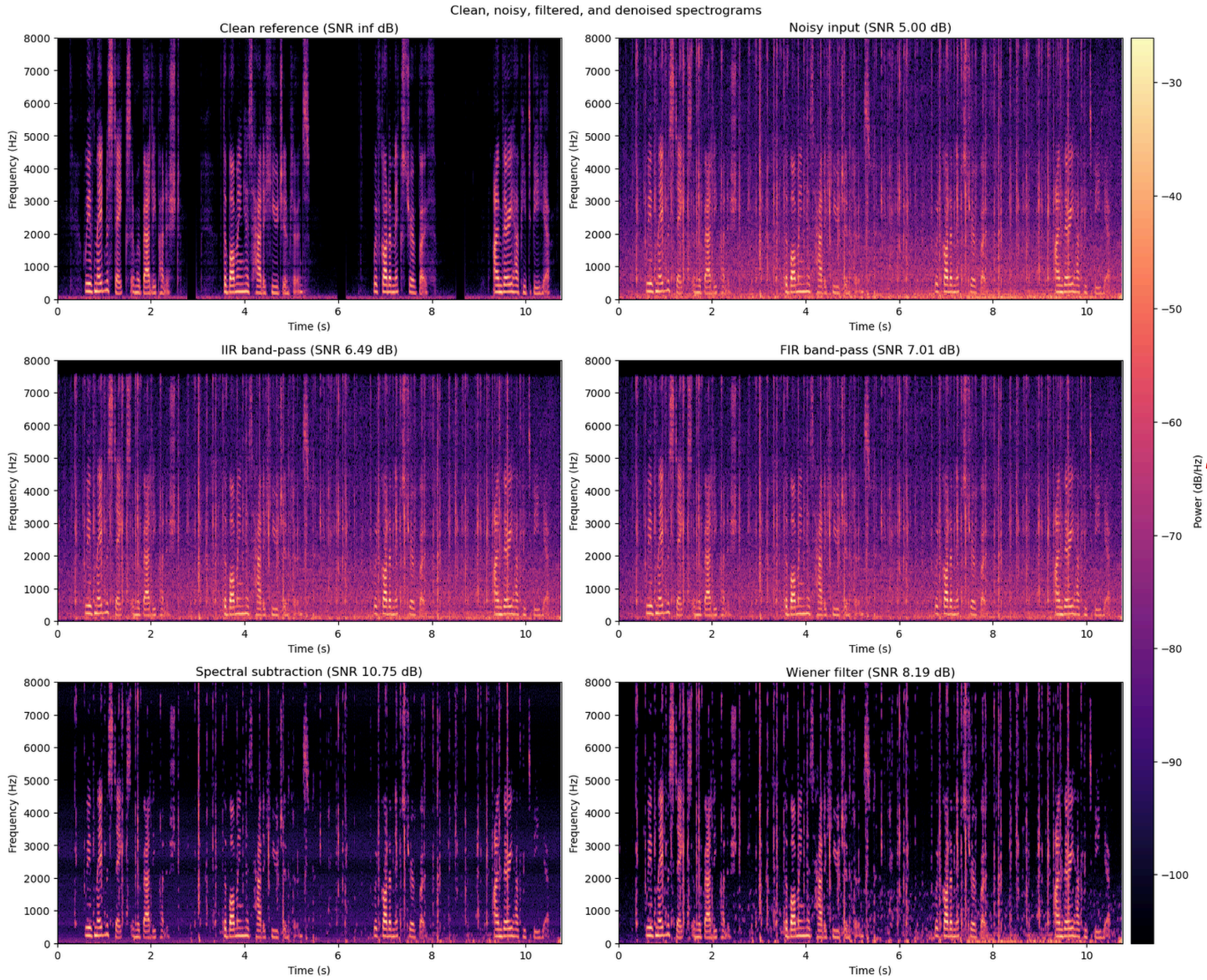


# FIR AND IIR BASELINE FILTER

```
Noise PSD power by candidate filter band:
  0-70 Hz: 0.000510092 (24.32% of estimated noise power)
  70-7500 Hz: 0.00145155 (69.21% of estimated noise power)
  7500-8000 Hz: 1.21106e-05 (0.58% of estimated noise power)
```



# STFT

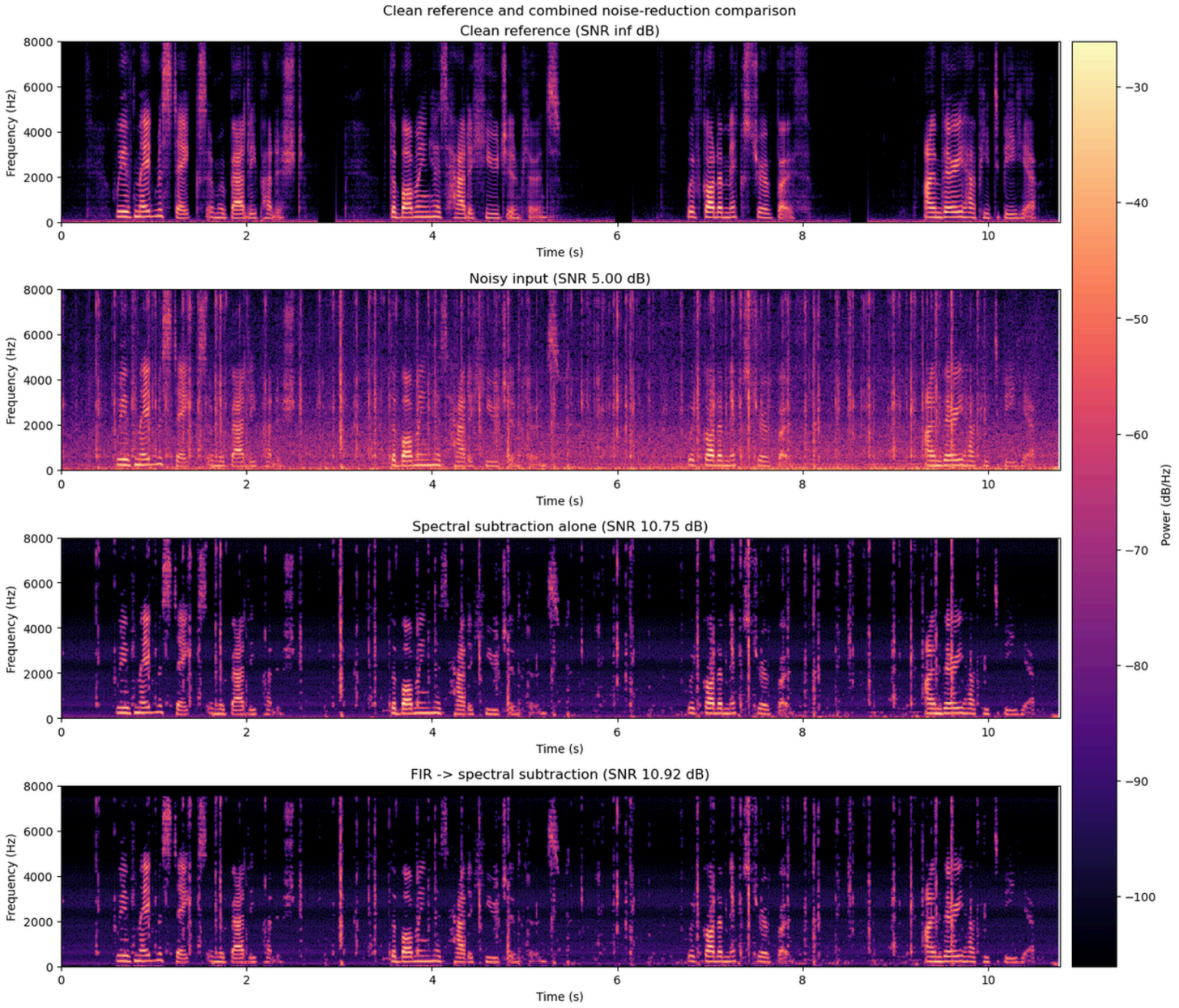





Measured input SNR: 5.00 dB

| Method | Output SNR | Improvement |
|---|---|---|
| Noisy input | 5.00 dB | +0.00 dB |
| IIR band-pass | 6.49 dB | +1.49 dB |
| FIR band-pass | 7.01 dB | +2.01 dB |
| Spectral subtraction (estimated profile) | 10.75 dB | +5.75 dB |
| Wiener filter | 8.19 dB | +3.19 dB |
| Oracle spectral subtraction | 14.45 dB | +9.45 dB |

# COMBINED DENOISING PIPELINE






Standalone FIR SNR: 7.01 dB
Standalone spectral subtraction SNR: 10.75 dB

| Combined pipeline | Output SNR | Improvement vs input |
|---|---|---|
| Noisy input | 5.00 dB | +0.00 dB |
| FIR -> spectral subtraction | 10.92 dB | +5.92 dB |
| IIR -> spectral subtraction | 10.83 dB | +5.83 dB |
| Oracle FIR -> spectral subtraction | 13.96 dB | +8.96 dB |

FIR -> spectral subtraction change vs standalone spectral subtraction: +0.17 dB

# TARGETED KEYBOARD CLICK EXPERIMENT



1. Find click:
   - Use the known noise track
   - Band-pass the audio
   - Use median over frame
   - Clean up the detections
2. Spectral subtraction
   - STFT
   - Noise profile
   - Subtract

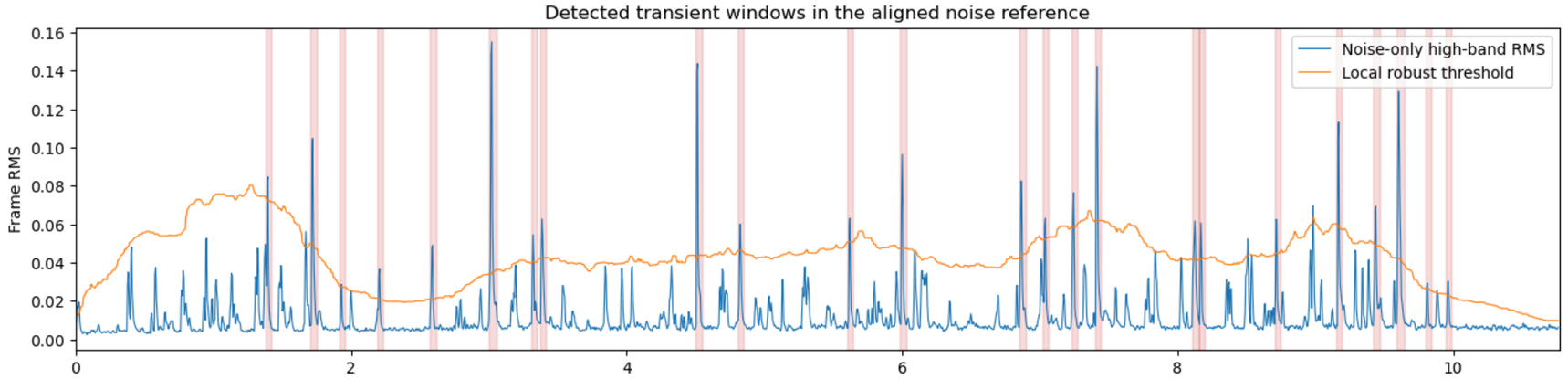
Goal:

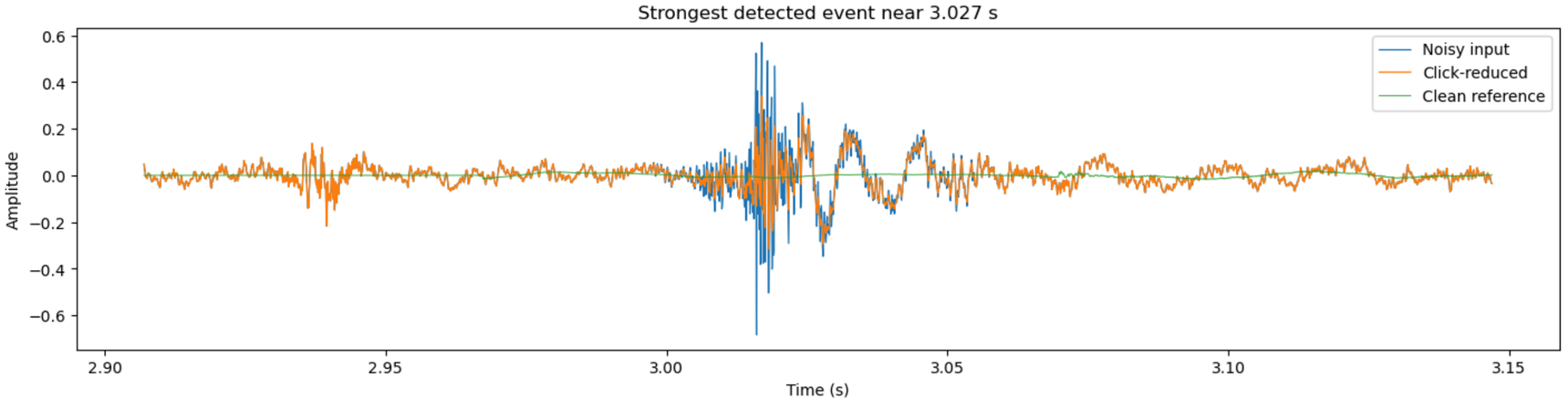
- removes clicks that would damage the spectral subtraction stage
- needs no clean signal or noise recording
- tells what kind of noise each file has

# TARGETED KEYBOARD CLICK EXPERIMENT

```
Detected transient events: 24
Total event-window coverage: 9.9% of the recording
Click attenuation target in 1500-7500 Hz: -12.0 dB
```

| Method | Overall SNR | Event-band reduction |
|---|---|---|
| Noisy input | 5.00 dB | 0.00 dB |
| Localized click suppression | 5.37 dB | 6.62 dB |
| Spectral subtraction | 10.75 dB | 3.19 dB |
| Click suppression -> spectral subtraction | 11.20 dB | 6.86 dB |
| Wiener filter | 8.19 dB | 0.66 dB |
| Click suppression -> Wiener filter | 8.87 dB | 7.22 dB |

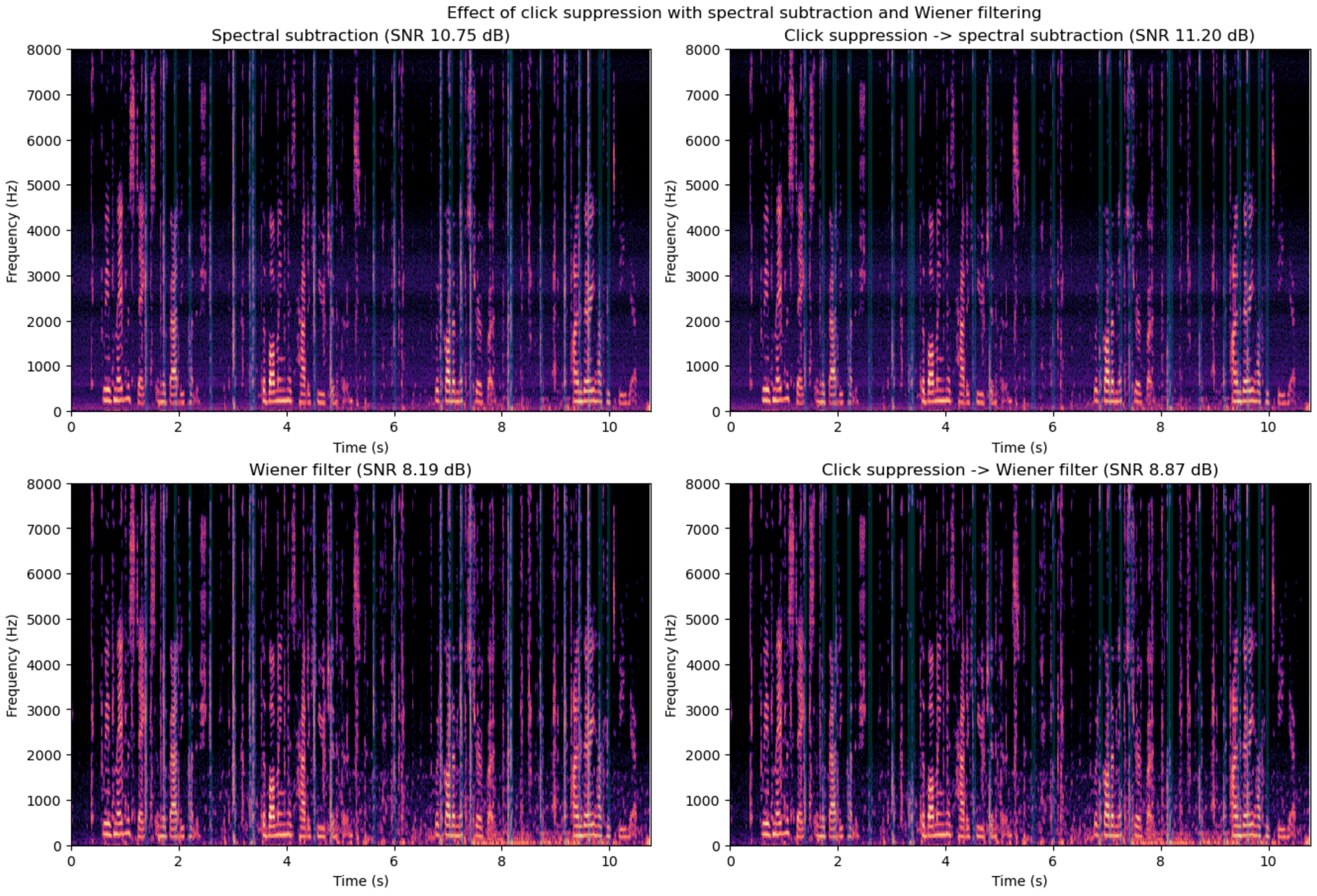


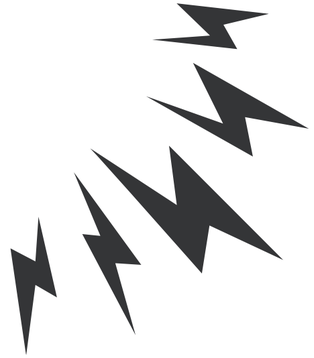



# TARGETED KEYBOARD CLICK EXPERIMENT





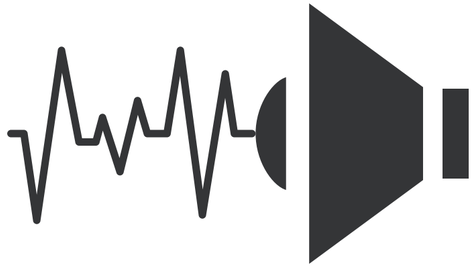






# TEST AND VALIDATE ALGORITHM WITH NOISY RECORDS

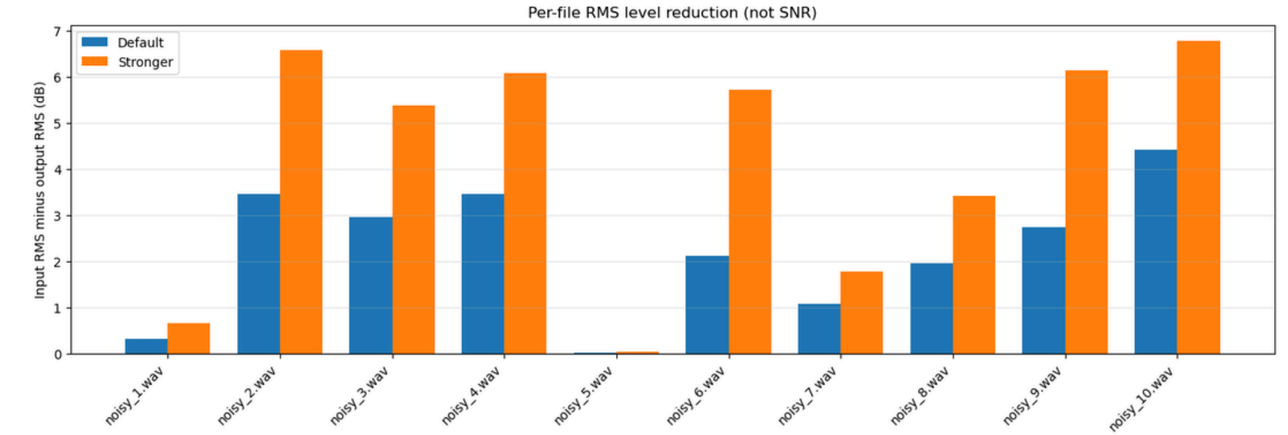


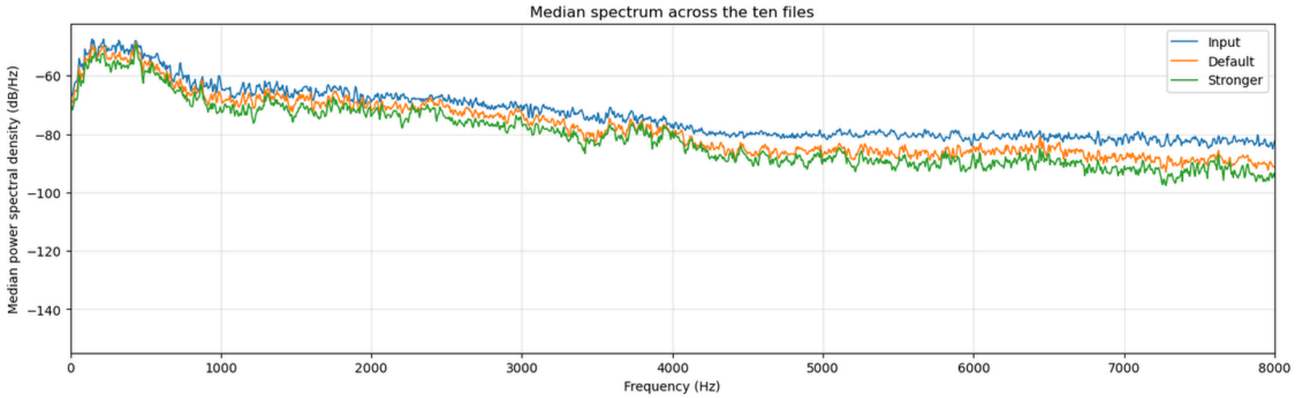


# VALIDATE ALGORITHM WITH 10 NOISY RECORDS

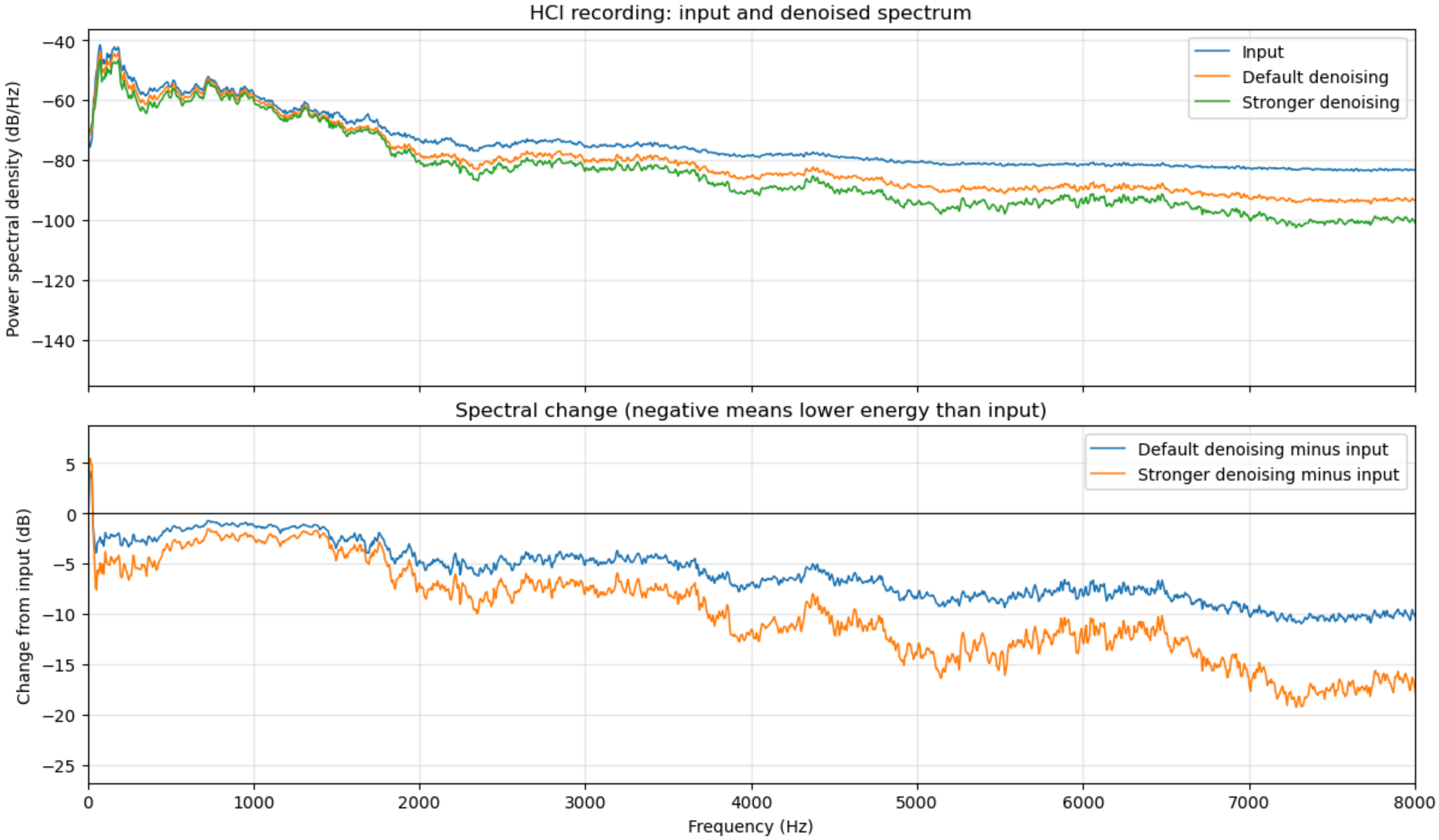
RMS changes are level differences, not SNR improvements.

| File | fs (Hz) | Duration | Input RMS | Default RMS | Change | Strong RMS | Change | Click candidates |
|---|---|---|---|---|---|---|---|---|
| noisy_1.wav | 22050 | 10.00s | -22.93 dBFS | -23.24 dBFS | -0.31 dB | -23.60 dBFS | -0.67 dB | 2/2 |
| noisy_2.wav | 22050 | 10.00s | -17.13 dBFS | -20.59 dBFS | -3.46 dB | -23.71 dBFS | -6.58 dB | 0/0 |
| noisy_3.wav | 22050 | 10.00s | -13.19 dBFS | -16.14 dBFS | -2.95 dB | -18.57 dBFS | -5.38 dB | 1/1 |
| noisy_4.wav | 22050 | 10.00s | -15.60 dBFS | -19.05 dBFS | -3.45 dB | -21.66 dBFS | -6.07 dB | 0/0 |
| noisy_5.wav | 22050 | 10.00s | -14.23 dBFS | -14.26 dBFS | -0.03 dB | -14.27 dBFS | -0.04 dB | 31/31 |
| noisy_6.wav | 22050 | 10.00s | -20.24 dBFS | -22.35 dBFS | -2.12 dB | -25.95 dBFS | -5.71 dB | 8/8 |
| noisy_7.wav | 22050 | 9.47s | -21.99 dBFS | -23.07 dBFS | -1.08 dB | -23.77 dBFS | -1.78 dB | 2/2 |
| noisy_8.wav | 22050 | 10.00s | -22.81 dBFS | -24.77 dBFS | -1.95 dB | -26.24 dBFS | -3.42 dB | 5/5 |
| noisy_9.wav | 22050 | 10.00s | -16.74 dBFS | -19.49 dBFS | -2.75 dB | -22.87 dBFS | -6.13 dB | 19/19 |
| noisy_10.wav | 22050 | 10.00s | -19.45 dBFS | -23.87 dBFS | -4.42 dB | -26.23 dBFS | -6.78 dB | 1/1 |

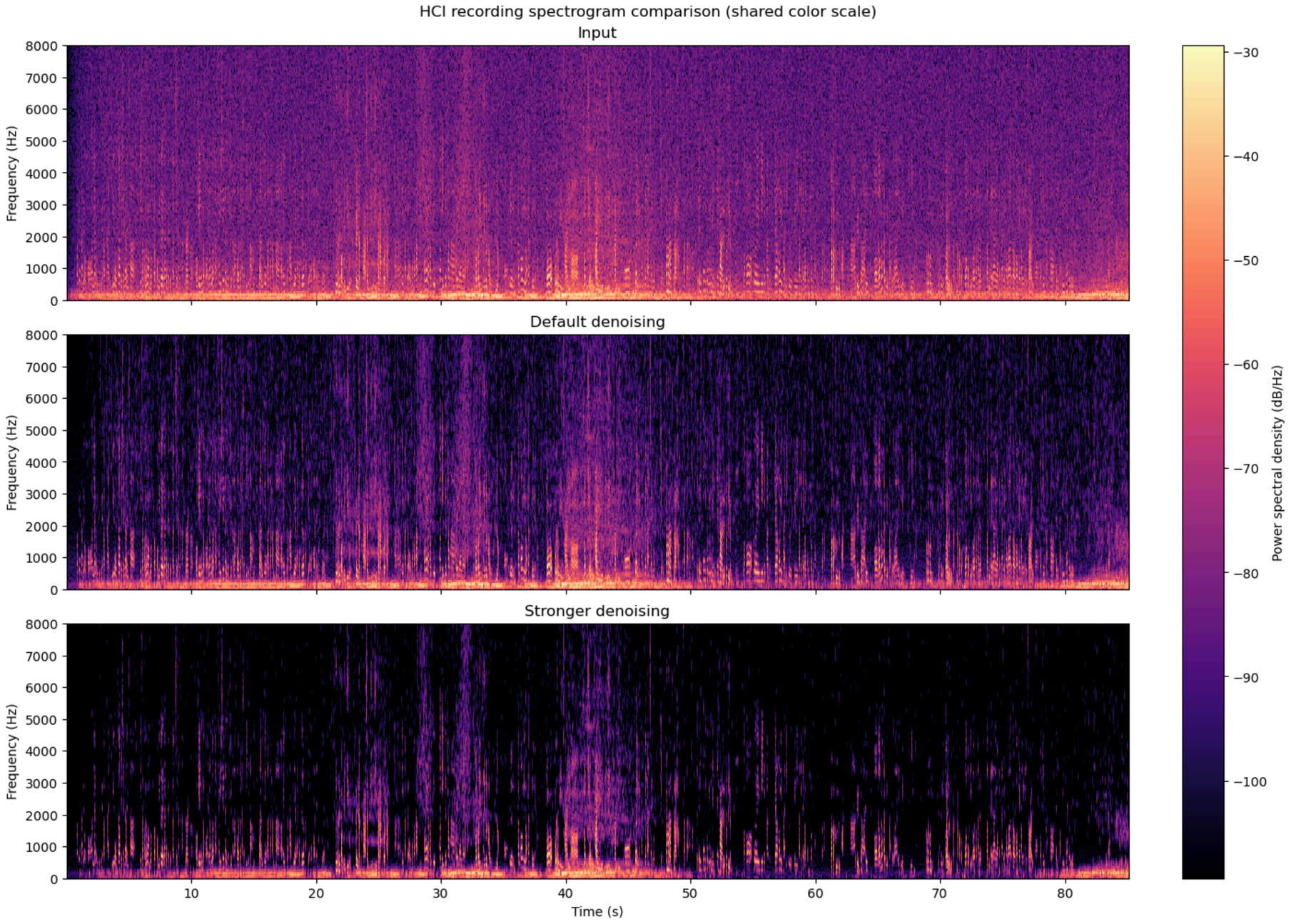


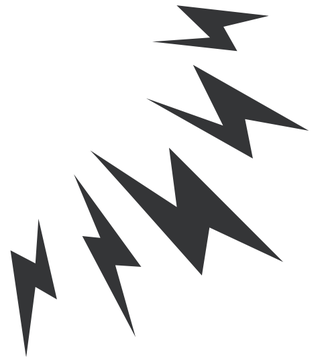



# TEST ALGORITHM WITH TARGET FILE



# TEST ALGORITHM WITH TARGET FILE

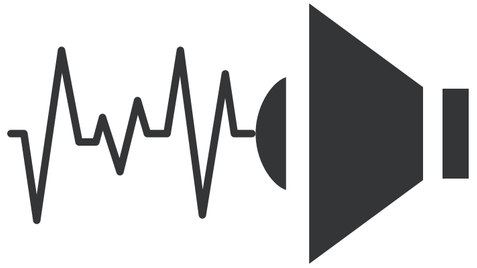






# FAILURE CASE

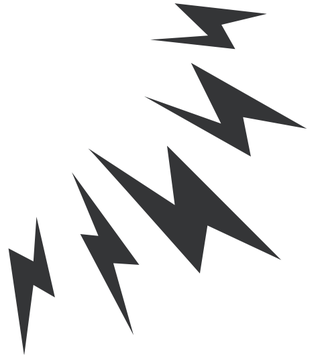




# FAILURE CASES



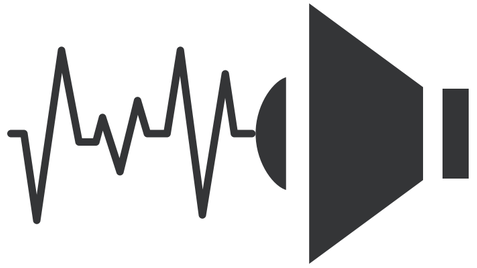
- **FIR/IIR filtering:** Limited when noise overlaps speech frequencies
- **Click detection:**May mistake speech transients for clicks and suppress wanted speech.
- **Aggressive denoising:** Can reduce speech quality along with noise.
- **Evaluation limitation:** No clean reference for the unknown and batch recordings, so noise-removal quality cannot be objectively confirmed.





# CONCLUSION

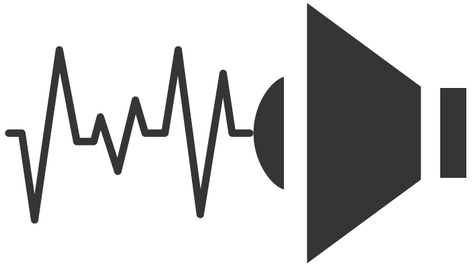




# CONCLUSION

- Spectral subtraction performed best in controlled testing (10.75 dB SNR)
- Click suppresson + Spectral subtraction : 11.20 dB
- Filter FIR/IIR can’t use in this noise reduction project
- Default click suppression plus spectral subtraction sounded best on the real recording, but is not yet effective enough for real world noise.
- Further improvement and testing are needed.







## MEMBER

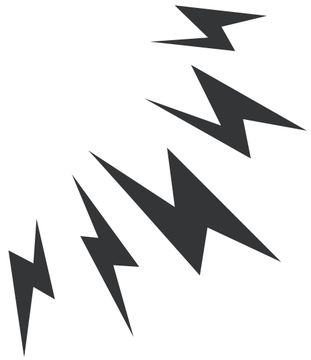
670710123 กัญญาณัฐ เขมนรากร
670710258 Sothea Sokea

## REFERENCE


- clean speech: https://www.kaggle.com/datasets/anhquocdevelop/clean-speech
- noisy audio: https://www.kaggle.com/datasets/367a72ddd73170fa1dee00402d00fcb5e0cbe08d84c0db192a69176bdf8606ee
- scipy : https://docs.scipy.org/doc/scipy/index.html
- Digital Signal Processing course lessons


## AI USAGE

- **Github Copilot:** helped edit and debug notebook code, explain DSP concepts, and document the project
- **Claude:** used to verify approaches and explain concepts or results.
- **ChatGPT:** used to check code and explanations.

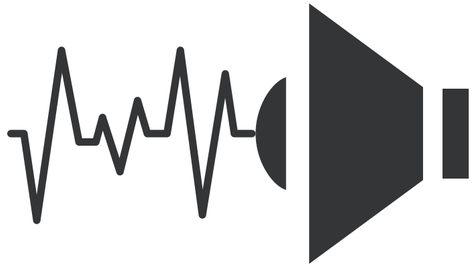


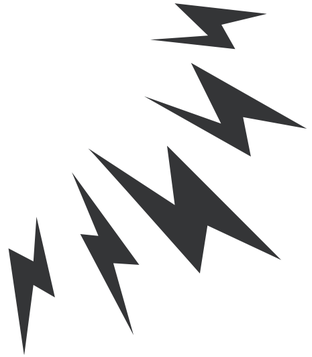
# MORE INFOMATION



github repository: audio_noise_reduction









# THANK YOU



